

Jay Niketan Pathare

jaypathare123@gmail.com | +1(716)303-9924 | NY, US | LinkedIn | GitHub

EDUCATION

Masters in Computer Science

University at Buffalo, The State University of New York, Buffalo, NY

August 2025 – December 2026

GPA: 3.852 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society's Institute of Technology, Mumbai University, India

August 2019 – May 2023

EXPERIENCE

Graduate Student Developer (CSE 611 Project) | Team of 5 | HeinOnline, Buffalo January 2026 – April 2026

- Engineered a hybrid OCR-LLM metadata extraction pipeline using Tesseract-OCR and Qwen-35B, feeding OCR output as context to the LLM per page to maximize extraction accuracy across 1,000+ legacy law journals.
- Automated end-to-end author metadata entry into a structured author database, reducing manual workload by 90% while sustaining 95%+ accuracy.
- Containerized the full pipeline using Docker for reproducible deployment and streamlined team collaboration.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records, and developed an IoT baby-warmer system with real-time PostgreSQL pipelines and 2s live monitoring.
- Developed a color-blindness diagnostic module achieving 97% clinical reliability, enabling automated report generation and real-time result visualization for medical practitioners.
- Containerized 3+ projects using Docker, reducing setup time by 90%, and integrated secure PDF generation and real-time video conferencing features.
- Collaborated with clients on UI/UX requirements, conducted 20+ technical interviews, mentored new hires and improved project delivery timelines.

Cloud Engineer Intern | Data Maven, Mumbai, India

November 2023 – May 2024

- Designed and deployed a scalable cloud infrastructure on AWS using EC2 and RDS, optimizing VPC networking to ensure secure, low-latency communication for backend services.
- Streamlined resource provisioning, reducing environment setup time and supporting the deployment of data-intensive applications.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Analyzed real-world datasets using Python (Pandas, NumPy) and MySQL to extract trends and actionable insights.

TECHNICAL SKILLS

- Languages:** Python, MySQL, PostgreSQL, HTML/CSS
- Frameworks:** Pandas, NumPy, Django, Matplotlib
- Artificial Intelligence & Machine Learning:** scikit-learn, PyTorch, TensorFlow, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Graph Neural Networks (GNNs)
- Software Engineering:** Object-Oriented Programming, REST API, Full-Stack Development, Backend Development, API Development, Unit Testing, ETL, LLM-based workflows, Prompt Engineering, AI-assisted automation
- Tools:** Docker, Git, Tableau, OAuth/Auth0, WebRTC

ACADEMIC PROJECTS

Fraud Detection using Graph Neural Networks | Team of 2 | University at Buffalo, NY

March 2026

- Built a GraphSAGE-based fraud detection model on the IEEE-CIS dataset, designed features and handled class imbalance for robust training and evaluation.
- Converted tabular transactions into a graph using shared card, device, and email features to capture relational patterns and improve detection accuracy.

Temperature Monitoring System | Team of 3 | University at Buffalo, NY

November 2025

- Architected an end-to-end IoT data pipeline, utilizing an Arduino Uno to stream continuous temperature readings over WiFi to a responsive ReactJS dashboard for live visualization.
- Developed a secure Django backend to process sensor data, integrating Auth0 to establish secure user authentication and safe data transmission.—Technologies: Arduino Uno, Django, ReactJS, Auth0.

Music Genre Classification | Team of 2 | VESIT, Mumbai, India

July 2022 – March 2023

- Classified audio files into genres using feature extraction techniques, achieving 97.68% accuracy.—Algorithms: CatBoost, KNN.

CERTIFICATIONS, AWARDS & PUBLICATIONS

- AWS Cloud Practitioner** July 2024
- Certificate of Appreciation:** Event Manager, CSE Workshop “GitHub: Hands-On from Basics to Advanced,” Dept. of Computer Science & Engineering, University at Buffalo. February 2026
- Research Publication:** J. Pathare, D. Ahuja, R. Singh, M. Sabnis, “Music Genre Classification,” International Journal of Research and Analytical Reviews (IJRAR), Volume 10, Issue 2, April 2023, ISSN: 2348-1269 Link.